TOPIC 02 · DATA ANALYTICS AND VISUALIZATION
LAB 8 THEME · CHARTS & VISUALISATION

Lab 8 — Build Charts That Tell the Hiring Story

Maps to	LO2: generate line, bar, column and combination charts with Gemini, preview and insert them, understand which charts update with the data, build linked charts and write titles that state the finding
Duration	30 minutes
Tools	Google Sheets · Gemini charts (Preview, Insert, Copy chart) · pivot table · Chart editor · OrchidBay_Hiring_Charts.xlsx
Mock data	data/OrchidBay_Hiring_Charts.xlsx
Course	Gemini for Google Sheet (C817) · v2.0
Prompts	https://tertiarycourses.github.io/C817-Gemini-for-Google-Sheet/prompts/#lab-8

Goal

Turn the 2026 hiring data into four clear charts - trend, sources, funnel, cost per hire - and know which ones update when the data changes.

What You Will Do

Now you show the findings from Lab 7. You will ask Gemini to create charts - a monthly trend line, hires by source and the hiring funnel - preview them and insert them. Each inserted chart lands on a new tab with its own copy of the data, so it does not update when Applications changes. You will prove that, then build a linked chart from a pivot table that does update. A cost-per-hire chart combines two tabs, and you will rewrite every title to state its finding and add alt text.

What You Will Build

OrchidBay_Hiring_Charts with four inserted Gemini charts, one linked chart built from a pivot table, finding-based titles and alt text

Prerequisites

- Lab 7 completed, or its Test It values to hand.
- This lab folder downloaded, including data/OrchidBay_Hiring_Charts.xlsx.

Data note

Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Steps

Step 1. Download OrchidBay_Hiring_Charts.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

PROMPT / CHECK — copy and adapt

```
Required state:
File : OrchidBay_Hiring_Charts (Google Sheets copy)
Tabs : About_This_Data | Applications (720 rows) | Jobs | Source_Costs | Prompt_Log
```

Step 2. Ask for a trend chart. Click Preview to see it, then Insert. Gemini adds a NEW tab with the chart and the data behind it - read the message at the bottom left. The line should peak in June (98).

PROMPT / CHECK — copy and adapt

```
INSERT:
Create a line chart with Month on the x-axis and the number of applications on the y-axis.
```

Step 3. Ask for two more charts and insert each. Hires by source, sorted: LinkedIn 11, Referral 10, JobStreet 8 ... Funnel: 720, 61, 47, 41.

PROMPT / CHECK — copy and adapt

```
INSERT:
Make a bar chart of hires (Stage is Hired) by Source, sorted from most to fewest.

INSERT:
Create a column chart of the hiring funnel: Applications (all rows), Interviewed (rows with an
Interview_Score), Offered (Stage is Offered or Hired) and Hired.
```

Step 4. Test whether the charts update. On Applications, change the Stage of one LinkedIn row from Rejected to Hired. Look at the hires-by-source chart: it does NOT change, because it is linked to the copy of the data on its own tab. Ask Gemini why, then undo your edit (Ctrl + Z / Cmd + Z).

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
Will the chart you inserted update if I change the Applications tab? How do I make a chart
that does?
```

Step 5. Build a linked chart. Ask Gemini for a pivot table, click Apply, then select the pivot table and choose Insert > Chart and pick a column chart. Repeat the Stage edit from Step 4: the Hired count for that month goes up by one in the pivot table and in this chart. Undo the edit again.

PROMPT / CHECK — copy and adapt

```
ACTION:
Create a pivot table on a new sheet that counts Candidate_ID with Month in rows and Stage in
columns.
```

Step 6. Combine two tabs in one chart. The lowest bar should be (S$125, Company Website) and Career Fair the highest (S$1,540).

PROMPT / CHECK — copy and adapt

```
INSERT:
Using the Applications and Source_Costs tabs, create a column chart of cost per hire by
Source, where cost per hire is applications x Cost_Per_Application divided by hires.
```

Step 7. Make every chart say something. Ask for finding-based titles, then double-click each chart > Customize > Chart & axis titles to apply them. Right-click each chart > Alt text and add one sentence. Optional: More options > Copy chart in the panel, and paste it into a Google Doc or Slides.

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
Suggest a title for each chart in this spreadsheet that states the main finding, not just the
measure.
For example: 'Applications peaked in June'.
```

Step 8. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

PROMPT / CHECK — copy and adapt

```
LOG:
List the prompts I sent you in this conversation. For each one, give a one-line summary of
what you did, so I can copy it into the Prompt_Log tab.
```

Sample Prompts

Use these prompts in the Gemini side panel in Google Sheets (or adapt them). Name the tab and columns, state the output, and always verify the result. The same prompts are in sample-prompts.txt in this lab folder.

Copy-ready prompts online: https://tertiarycourses.github.io/C817-Gemini-for-Google-Sheet/prompts/#lab-8

#	Purpose	Prompt
1	Pie	Create a pie chart of applications by Source.
2	Stacked	Create a stacked column chart of applications by Month, split by Stage.
3	Combo	Create a chart with applications as columns and hires as a line, by Month.
4	Help	What chart type is best to show the share of each source over time?
5	Graph	Create a graph that shows the percentage of applications that came from each Position.

Test It

Check	Expected result
Trend chart	Line chart on its own tab; peak June 98.
Source and funnel	Bars sorted; LinkedIn 11 first. Funnel 720 / 61 / 47 / 41.
Static vs linked	Inserted chart did not change after the Stage edit; the Month x Stage pivot chart did.
Cost per hire	Company Website S$125 lowest; Career Fair S$1,540 highest.
Titles and alt text	Every chart title states a finding; every chart has alt text.

Checkpoint

Topic 2 ends here. Lab 9 starts the interactive dashboard in its own workbook.

Troubleshooting

If this happens	Fix
No Preview or Insert button.	Ask again with the word 'chart' and the chart type. Make sure you are in the Google Sheets copy.
The chart counts the header as a month.	Ask Gemini to use data rows only, or edit the data range in the Chart editor.
Gemini put the chart on a new tab, but you wanted it on Applications.	That is how Insert works. Cut the chart (three dots > Cut) and paste it on another tab, or build it with Insert > Chart.
You can't find Copy chart.	In the Gemini panel, open More options (three dots) on the chart response.

Challenge

Build a combo chart from the Lab 8 pivot table that shows applications (columns) and hire rate (line on a second axis).

Reflection

When is a static chart fine, and when must a chart be linked to live data?